

5.1 Colour and perception ↕

□ Read

The following reading offers a light-hearted insight into the daily routines of great minds through history.

Skim read the article and consider:

- How would you approach creating a visualisation of your daily routine?
- What techniques would you use, and why?

Ensure that you can justify your thinking and bring your answer to the workshop.

Burkeman, O. (2013, October 5). [Rise and shine: the daily routines of history's most creative minds](https://www.theguardian.com/science/2013/oct/05/daily-rituals-creative-minds-mason-currey). ↗ (<https://www.theguardian.com/science/2013/oct/05/daily-rituals-creative-minds-mason-currey>) *The Guardian*.

Colour

Colour can evoke emotions, cause reactions, create moods and attract attention, which means that it can be used effectively to communicate key aspects of a chart or table. We can use it more effectively if we understand how it is perceived. Without this understanding, colour can be misused and its misuse can lead to confusion.

□ Watch: Purposeful use of colour – part 1 [17.31 mins]

Watch the following video, which explains why colour is a powerful tool when used well. It also describes the three attributes of colour, explains how colour is perceived, and sets out the difference between complementary and analogous colours.

This is useful as it provides a fundamental understanding of how colour can enhance or detract from the message in a visualisation.



[Transcript](#)

<https://learn.usc.edu.au/courses/35242/files/3138642?wrap=1>

Source: © University of the Sunshine Coast (2023).

Download the following slides for note taking: [ICT702 Purposeful use of colour – part 1 \(https://learn.usc.edu.au/courses/35242/files/3138661?wrap=1\)](https://learn.usc.edu.au/courses/35242/files/3138661?wrap=1)

Attributes of colour

Colour has three attributes: hue, saturation and luminance.

- *Hue* is the base of a colour. There are three primary hues (red, yellow and blue), and it is not possible to mix or form a primary hue by combining other hues. Combinations of the primary hues form secondary colours, such as orange, green and violet, and a variety of tertiary colours.
- *Saturation* is the amount of grey in a colour. Saturation determines the intensity or purity of the hue in the colour. An absolutely pure hue has no greyness – it is 100% saturated. All hues become grey at 0% saturation.
- *Luminance* is the relative degree of white or black within a colour. As you would expect, adding white to a hue results in a lighter colour and adding black results in a darker colour. Contrast can be achieved by using differences in the luminance of a colour. Therefore, luminance can be used to create hierarchy. All hues become white at 100% luminance and black at 0% luminance.

Colour psychology and colour symbolism

Colour psychology refers to the innate relationships between colour and human behaviour. Cool hues are thought to be calming, reassuring and soothing, whereas warm hues are considered to invoke feelings of danger, energy and passion. *Colour symbolism* relates to the cultural meanings and significance associated with colour. Some other factors to consider when using colour in visuals are:

- *perceived colour*. The perception of colour is not absolute. Colour is perceived differently depending on what other colours are being viewed at the same time around the object at the centre of the field of vision. In other words, the colour of the object is perceived in contrast to any adjacent colours.
- *complementary colours*. These create what is called colour dissonance. This means that the colours appear stark and there is strong contrast between them. This is useful when you want objects to stand out, but their overuse can be distracting.
- *analogous colours*. These colours are directly adjacent on a colour wheel. Therefore, they have underlying similarities in their hues and appear softer and smoother when used together. There is less contrast between them. However, their overuse can still be distracting.

□ Read

The following three articles by Jeff Desjardins consider colour within business and branding.

1. Desjardins, J. (2016, June 20). [The psychology of color in business](https://www.visualcapitalist.com/psychology-of-color-business/)  [\(https://www.visualcapitalist.com/psychology-of-color-business/\)](https://www.visualcapitalist.com/psychology-of-color-business/). *Visual Capitalist*.

As you read the first article, consider and make notes on the ease with which the infographic can be understood.

2. Desjardins, J. (2016, August 25). [The psychology behind logo and colour choice](https://www.visualcapitalist.com/psychology-behind-logo-color-choice/)  [\(https://www.visualcapitalist.com/psychology-behind-logo-color-choice/\)](https://www.visualcapitalist.com/psychology-behind-logo-color-choice/). *Visual Capitalist*.

As you read the second article, consider and make notes on the importance of colour in business.

3. Desjardins, J. (2017, February 20). [Color in branding: what does it say about your industry?](https://www.visualcapitalist.com/color-in-branding-industry/)  [\(https://www.visualcapitalist.com/color-in-branding-industry/\)](https://www.visualcapitalist.com/color-in-branding-industry/) *Visual Capitalist*.

As you read the third article, consider and make notes on:

- the ease with which the infographic can be understood
- how you would approach creating a visualisation of the data
- what techniques you would use and why.

□ Watch: The science of colour perception [4.54 mins]

This video is useful as it provides an explanation of what colours are and how our eyes and brains perceive them.

As you watch consider:

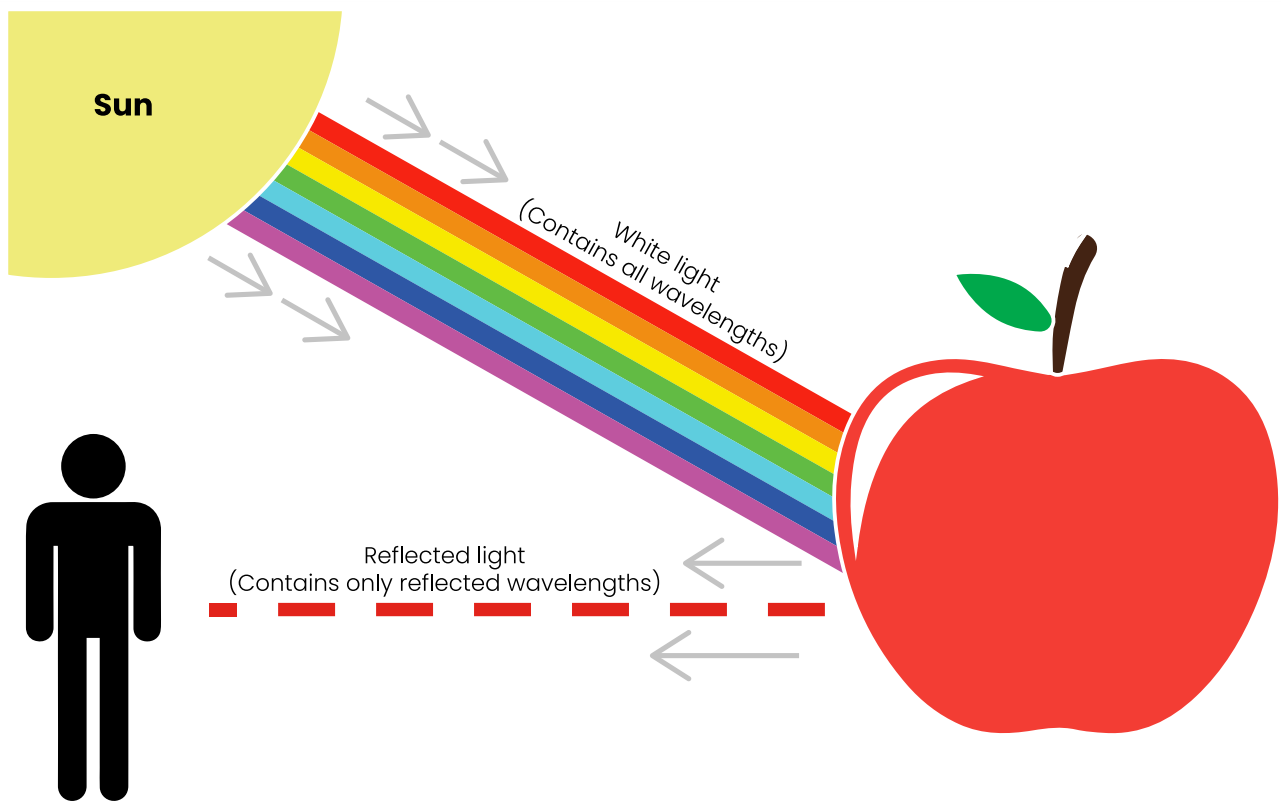
- How does this understanding influence your use of colour?
- Why is this important when creating useful visualisations?



Source: © Calder Ruhl Hansen (2016).

Light plays a crucial role in our visual experiences. It can enhance or diminish the appeal of different objects and spaces. The interplay between light and shadows, colour temperature, intensity, and direction have an effect the aesthetics and mood of images we see. Figure 4.1 below shows how the colours we see are the result of the light reflected off an object. When light interacts with an object, the object absorbs certain wavelengths of natural light and reflects or transmits others, giving it a specific colour.

Figure 4.1 The science of colour perception



Source: © ARTZ 363 (2015).

[Transcript \(https://learn.usc.edu.au/courses/35242/files/3138643?wrap=1\)](https://learn.usc.edu.au/courses/35242/files/3138643?wrap=1)

It is important to consider light in data visualisation, as light helps viewers to distinguish between different colours representing data points, categories and variables. Proper lighting helps prevent colour confusion and enhances clarity in interpreting visualised information.